

Unit 1. Vectors, Matrices, and Linear Systems

I. Vectors

I.1–2. Coordinates and displacements

A vector in $\mathbb{R}^n$ is an ordered list $x = (x_1, \dots, x_n)$ of real numbers. Two vectors are equal when their corresponding coordinates are equal. For $x, y \in \mathbb{R}^n$ and $c \in \mathbb{R}$, define

$$x + y = (x_1 + y_1, \dots, x_n + y_n), \quad cx = (cx_1, \dots, cx_n).$$

The zero vector is $0 = (0, \dots, 0)$, and $x - y = x + (-1)y$. These operations satisfy

$$\begin{aligned} x + y &= y + x, & (x + y) + z &= x + (y + z), \\ c(x + y) &= cx + cy, & (c + d)x &= cx + dx, \\ c(dx) &= (cd)x, & 1x &= x, \\ x + 0 &= x, & x + (-x) &= 0. \end{aligned}$$

Each identity follows by applying the corresponding real-number identity to every coordinate. A **linear combination** of vectors $v_1, \dots, v_k$ is a vector $c_1v_1 + \dots + c_kv_k$, with real coefficients.

The same coordinates can specify a point. The displacement from a point p to a point q is $q - p$; translating both points by a leaves it unchanged:

$$(q + a) - (p + a) = q - p.$$

I.3. Scalar product

The **scalar product**, or **dot product**, of $x, y \in \mathbb{R}^n$ is

$$x \cdot y = \sum_{i=1}^n x_i y_i.$$

For example, $(1, 3, -2) \cdot (-1, 4, -3) = -1 + 12 + 6 = 17$. For $x, y, z \in \mathbb{R}^n$ and $c, d \in \mathbb{R}$,

$$\begin{aligned} x \cdot y &= y \cdot x, \\ x \cdot (cy + dz) &= c(x \cdot y) + d(x \cdot z), \\ x \cdot x &\geq 0, \quad x \cdot x = 0 \iff x = 0. \end{aligned}$$

Symmetry follows from $x_i y_i = y_i x_i$. To verify the second identity, expand the coordinates before collecting the two sums:

$$x \cdot (cy + dz) = \sum_i x_i (cy_i + dz_i) = c \sum_i x_i y_i + d \sum_i x_i z_i = c(x \cdot y) + d(x \cdot z).$$

Symmetry gives the same rule in the first argument. Finally, $x \cdot x = \sum_i x_i^2$. Every summand is nonnegative; if $x \neq 0$, some $x_j \neq 0$, so the sum is at least $x_j^2 > 0$.

Vectors x, y are **orthogonal**, written $x \perp y$, if $x \cdot y = 0$. The zero vector is orthogonal to every vector. Let e_i have coordinate 1 in position i and 0 elsewhere. Then

$$x = \sum_{i=1}^n x_i e_i, \quad x \cdot e_i = x_i, \quad e_i \cdot e_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

I.4. Norm, projection, and angles

The **norm** of x and the **distance** between x and y are

$$\|x\| = \sqrt{x \cdot x}, \quad d(x, y) = \|x - y\|.$$

In the plane, the classical Pythagorean theorem identifies $\sqrt{x_1^2 + x_2^2}$ with the length of the vector. In space, applying it first in the coordinate plane and then in the perpendicular direction gives $\sqrt{x_1^2 + x_2^2 + x_3^2}$. The norm extends this length formula to $\mathbb{R}^n$. Thus $\|x\| \geq 0$, with equality exactly when $x = 0$, and

$$\|cx\| = \sqrt{c^2(x \cdot x)} = |c| \|x\|.$$

A vector has **unit norm** if its norm is 1. For $x \neq 0$,

$$\left\| \frac{x}{\|x\|} \right\| = \frac{1}{\|x\|} \|x\| = 1.$$

Two nonzero vectors have the **same direction** when one is a positive multiple of the other. Thus $x/\|x\|$ is the unit vector in the direction of x , and $x = \|x\| (x/\|x\|)$.

For a center a and radius $r > 0$, the sets

$$\{x : \|x - a\| < r\}, \quad \{x : \|x - a\| \leq r\}, \quad \{x : \|x - a\| = r\}$$

are the **open ball**, **closed ball**, and **sphere**. Every point of the sphere is $a + ru$ for a unit vector u : one direction follows from $\|ru\| = r$, and the other from $u = (x - a)/r$. In $\mathbb{R}^2$ the sphere is a circle.

Proposition 1.1 (Pythagoras). For $x, y \in \mathbb{R}^n$,

$$\|x + y\|^2 = \|x\|^2 + 2x \cdot y + \|y\|^2.$$

Consequently, $x \perp y$ if and only if $\|x + y\|^2 = \|x\|^2 + \|y\|^2$.

Proof. Using the scalar-product identities,

$$\begin{aligned} \|x + y\|^2 &= (x + y) \cdot (x + y) \\ &= x \cdot x + x \cdot y + y \cdot x + y \cdot y \\ &= \|x\|^2 + 2x \cdot y + \|y\|^2. \end{aligned}$$

Thus the norm identity without the cross term holds exactly when $2x \cdot y = 0$, which is the definition of orthogonality. $\square$

Proposition 1.2 (Projection onto a line). Let $b \neq 0$. For each $x \in \mathbb{R}^n$, there is a unique real number c such that $x - cb \perp b$, namely

$$c = \frac{x \cdot b}{b \cdot b}.$$

The vector $p = cb$ is the **orthogonal projection** of x onto the line of multiples of b . Writing $r = x - p$,

$$x = p + r, \quad p \perp r, \quad \|x\|^2 = \|p\|^2 + \|r\|^2.$$

Proof. We seek a multiple cb whose difference from x is perpendicular to b . The condition is

$$0 = (x - cb) \cdot b = x \cdot b - c(b \cdot b).$$

Since $b \cdot b > 0$, this equation forces the displayed value of c . Conversely, that value gives

$$\left(x - \frac{x \cdot b}{b \cdot b} b \right) \cdot b = x \cdot b - \frac{x \cdot b}{b \cdot b} (b \cdot b) = 0.$$

Hence the required multiple exists and is unique. With $p = cb$ and $r = x - cb$, we have $p \cdot r = c(b \cdot r) = 0$, so Pythagoras applies to $x = p + r$. $\square$

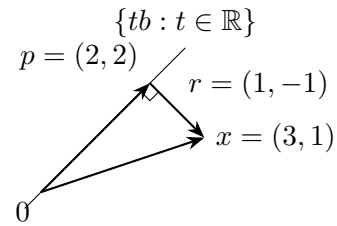
The coefficient c is a scalar; the projection cb is a vector. When b has unit norm, $c = x \cdot b$.

Example 1.1.

For $x = (3, 1)$ and $b = (1, 1)$,

$$c = \frac{3+1}{1+1} = 2, \quad p = (2, 2), \quad r = (1, -1).$$

Thus $r \cdot b = 0$, and Pythagoras gives $\|x\|^2 = 10 = 8 + 2 = \|p\|^2 + \|r\|^2$.



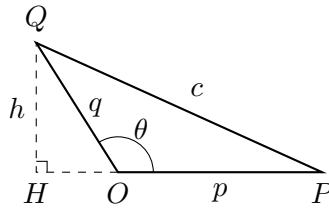
The geometric angle. For two nonzero vectors in the plane or in space, place their tails at the same point. Their geometric angle θ lies in $[0, \pi]$. The following proof uses ordinary Euclidean lengths and right-triangle trigonometry.

Proposition 1.3 (Cosine law). If two sides of a triangle have lengths $p, q > 0$, their included angle is θ , and the opposite side has length c , then

$$c^2 = p^2 + q^2 - 2pq \cos \theta.$$

The formula also holds for a degenerate triangle with $\theta = 0$ or $\theta = \pi$.

Proof. Let the vertices be O, P, Q , with $|OP| = p$, $|OQ| = q$, and $\angle POQ = \theta$. For $0 < \theta < \pi$, drop a perpendicular from Q to the line through O, P , meeting it at H . Let t be the signed distance from O to H , positive in the direction of P , and put $h = |HQ|$.



For an acute angle, $t = q \cos \theta$. For a right angle, both sides are zero. For an obtuse angle, H lies on the opposite ray and $t = -q \cos(\pi - \theta) = q \cos \theta$. The classical geometric Pythagorean theorem gives $t^2 + h^2 = q^2$ and

$$c^2 = (p - t)^2 + h^2 = p^2 - 2pt + (t^2 + h^2) = p^2 + q^2 - 2pq \cos \theta.$$

The squared distance $(p - t)^2$ also covers a foot beyond P . If $\theta = 0$, then $c = |p - q|$; if $\theta = \pi$, then $c = p + q$. Substitution proves the two degenerate cases. $\square$

Apply the cosine law to the triangle with vertices $0, a, b$, for nonzero $a, b \in \mathbb{R}^2$ or $\mathbb{R}^3$. Euclidean distance agrees with the coordinate norm by Pythagoras, so

$$\|a - b\|^2 = \|a\|^2 + \|b\|^2 - 2\|a\|\|b\|\cos \theta.$$

On the other hand, expansion of the coordinate dot product gives

$$\|a - b\|^2 = (a - b) \cdot (a - b) = \|a\|^2 + \|b\|^2 - 2a \cdot b.$$

Comparing the two expressions yields the geometric formula

$$\boxed{a \cdot b = \|a\|\|b\|\cos \theta.}$$

Thus the coordinate definition of the dot product agrees with the geometric definition $\|a\|\|b\|\cos \theta$. For nonzero vectors, $a \cdot b = 0$ is equivalent to $\theta = \pi/2$.

The coordinate bound $|x_i| \leq \|x\|$ follows from $x_i^2 \leq \sum_j x_j^2$. Projection gives the same bound along any unit direction e . Put $c = x \cdot e$. Then

$$x = (x - ce) + ce, \quad \|x\|^2 = \|x - ce\|^2 + c^2 \geq c^2,$$

so $|x \cdot e| \leq \|x\|$. The next theorem allows an arbitrary second vector.

Theorem 1.4 (Schwarz inequality). For all $x, y \in \mathbb{R}^n$,

$$|x \cdot y| \leq \|x\| \|y\|.$$

If $y \neq 0$, equality holds exactly when x is a scalar multiple of y . If $y = 0$, equality always holds.

Proof. The case $y = 0$ gives $0 = 0$. For $y \neq 0$, let $c = (x \cdot y) / \|y\|^2$ and $r = x - cy$. By Proposition 1.2,

$$\|x\|^2 = \|r\|^2 + c^2 \|y\|^2 = \|r\|^2 + \frac{(x \cdot y)^2}{\|y\|^2} \geq \frac{(x \cdot y)^2}{\|y\|^2}.$$

Multiply by $\|y\|^2$ and take nonnegative square roots. Equality holds exactly when $\|r\| = 0$, equivalently $x = cy$. $\square$

For nonzero $a, b \in \mathbb{R}^n$, Schwarz gives

$$-1 \leq \frac{a \cdot b}{\|a\| \|b\|} \leq 1.$$

We therefore **define the angle** between them to be the unique $\theta \in [0, \pi]$ such that

$$\cos \theta = \frac{a \cdot b}{\|a\| \|b\|}, \quad \text{equivalently} \quad \theta = \arccos \frac{a \cdot b}{\|a\| \|b\|}.$$

This extends the geometric angle in $\mathbb{R}^2$ and $\mathbb{R}^3$. An angle with the zero vector is undefined. For nonzero vectors, a positive, zero, or negative dot product means respectively $\theta < \pi/2$, $\theta = \pi/2$, or $\theta > \pi/2$.

Proposition 1.5 (Triangle inequality). For $x, y \in \mathbb{R}^n$, $\|x + y\| \leq \|x\| + \|y\|$.

Proof. Expanding and applying Schwarz gives

$$\|x + y\|^2 = \|x\|^2 + 2x \cdot y + \|y\|^2 \leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2 = (\|x\| + \|y\|)^2.$$

Both sides of the claimed inequality are nonnegative, so taking square roots preserves the inequality. $\square$

I.5. Parametric lines

The line through $a \in \mathbb{R}^n$ with direction $d \neq 0$ is

$$\ell = \{a + td : t \in \mathbb{R}\}.$$

For distinct points a, q , take $d = q - a$. Restricting $0 \leq t \leq 1$ gives the segment joining the two points:

$$a + t(q - a) = (1 - t)a + tq, \quad a + 0(q - a) = a, \quad a + 1(q - a) = q.$$

The distance traveled from a is $\|t(q - a)\| = t\|q - a\|$, so the parameter is the fraction of the segment traversed. Differences of points on ℓ are multiples of d :

$$(a + td) - (a + sd) = (t - s)d.$$

Replacing d by cd , for $c \neq 0$, gives the same line because ct still ranges over all real numbers. If $c < 0$, increasing the new parameter moves in the opposite direction.

For $a = (1, -3, 4)$, $q = (5, 1, -2)$, the point one third of the way from a to q is

$$a + \frac{1}{3}(q - a) = (1, -3, 4) + \frac{1}{3}(4, 4, -6) = (7/3, -5/3, 2).$$

Example 1.2. Take $a = (1, 0, 1)$ and $d = (1, 1, 0)$. Then

$$x = a + td = (1 + t, t, 1) \iff \begin{cases} x_1 - x_2 = 1, \\ x_3 = 1. \end{cases}$$

The parameterization implies both equations. Conversely, if the equations hold, setting $t = x_2$ recovers the parameterization.

I.6. Planes

A plane through $a \in \mathbb{R}^3$, with nonzero **normal vector** n , has equation

$$n \cdot (x - a) = 0, \quad \text{equivalently} \quad n \cdot x = n \cdot a.$$

If p, q lie in the plane, then $n \cdot (q - p) = 0$: the normal is perpendicular to displacements within the plane. A position vector p need not be perpendicular to n . For $n = (n_1, n_2, n_3)$, the equation reads

$$n_1x_1 + n_2x_2 + n_3x_3 = n \cdot a.$$

Thus the coefficients of a plane equation give its normal. Multiplying the equation by any nonzero scalar changes neither its solutions nor the line of normal vectors; a negative scalar reverses the normal. In $\mathbb{R}^2$, the equation $n \cdot (x - a) = 0$ defines the line through a perpendicular to n .

More generally, for $n \in \mathbb{R}^k \setminus \{0\}$, the equation $n \cdot x = \alpha$ defines a **hyperplane**. It contains the origin exactly when $\alpha = 0$. If a is one solution, then

$$\{x : n \cdot x = \alpha\} = \{a + v : n \cdot v = 0\}.$$

This follows by substituting $x = a + v$ and subtracting $n \cdot a = \alpha$.

Example 1.3. The plane with normal $n = (1, -1, 0)$ through $a = (1, 0, 1)$ is

$$x_1 - x_2 = 1.$$

It contains the line in Example 1.2, since $n \cdot d = 0$. The point $(1, 0, 0)$ is in the plane but not on that line: the line also requires $x_3 = 1$.

Example 1.4 (A plane through three points). Find a plane through $p_1 = (1, 2, -1)$, $p_2 = (-1, 1, 4)$, and $p_3 = (1, 3, -2)$. A normal $n = (a, b, c)$ must satisfy

$$n \cdot (p_2 - p_1) = 0, \quad n \cdot (p_3 - p_1) = 0.$$

Since $p_2 - p_1 = (-2, -1, 5)$ and $p_3 - p_1 = (0, 1, -1)$, these are

$$-2a - b + 5c = 0, \quad b - c = 0.$$

Taking $b = c = 1$ gives $a = 2$. Hence $n = (2, 1, 1)$ is a nonzero normal, and the plane is

$$2x_1 + x_2 + x_3 = n \cdot p_1 = 3.$$

Substituting each of the three points gives 3.

Two planes are parallel when their normals are parallel, and perpendicular when their normals have dot product zero. Choosing normals n, m determines an angle by $\cos \theta = (n \cdot m) / (\|n\| \|m\|)$. Reversing one normal replaces θ by $\pi - \theta$. The smaller, unoriented angle $\phi \in [0, \pi/2]$ therefore satisfies

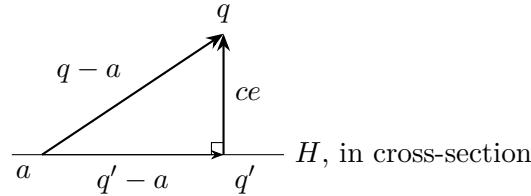
$$\cos \phi = \frac{|n \cdot m|}{\|n\| \|m\|}.$$

For $2x_1 - x_2 + x_3 = 0$ and $x_1 + 2x_2 - x_3 = 1$, the displayed normals are $(2, -1, 1)$, $(1, 2, -1)$. Their dot product is -1 and both have norm $\sqrt{6}$; thus $\cos \theta = -1/6$ and $\cos \phi = 1/6$.

Proposition 1.6 (Distance from a point to a plane). Let $H = \{x : n \cdot (x - a) = 0\}$, where $n \neq 0$. For a point q , the perpendicular foot on H and the distance to H are

$$q' = q - \frac{(q - a) \cdot n}{\|n\|^2} n, \quad \text{dist}(q, H) = \frac{|(q - a) \cdot n|}{\|n\|}.$$

Proof. Set $e = n / \|n\|$, a unit normal. The component of $q - a$ along e is $c = (q - a) \cdot e$. Subtracting this normal component gives the candidate foot $q' = q - ce$.



Indeed,

$$(q' - a) \cdot e = (q - a) \cdot e - c(e \cdot e) = c - c = 0,$$

so $q' \in H$, and $q - q' = ce$ is perpendicular to the plane. For any $x \in H$, the displacement $q' - x$ is perpendicular to e . Therefore

$$q - x = ce + (q' - x), \quad \|q - x\|^2 = c^2 + \|q' - x\|^2 \geq c^2.$$

Equality holds exactly at $x = q'$. The shortest distance is consequently

$$\|ce\| = |c| = \frac{|(q - a) \cdot n|}{\|n\|}.$$

□

Example 1.5. For $q = (1, 3, 5)$ and the plane $-x_1 + x_2 - x_3 = -5$, take $a = (-1, 1, 7)$, $n = (-1, 1, -1)$. Then

$$q - a = (2, 2, -2), \quad (q - a) \cdot n = 2, \quad \|n\| = \sqrt{3}.$$

The distance is $2/\sqrt{3}$, and the foot is

$$q' = q - \frac{2}{3}n = (5/3, 7/3, 17/3).$$

Its coordinates satisfy $-5/3 + 7/3 - 17/3 = -5$, as required.

II. Matrices and linear equations

II.1. Matrices

An $m \times n$ matrix $A = (a_{ij})$ has m rows and n columns; a_{ij} is the entry in row i , column j . Two matrices are equal when they have the same size and equal corresponding entries. For matrices of the same size,

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = ca_{ij}.$$

The zero matrix has every entry zero. A matrix is **square** when $m = n$. Addition and scalar multiplication satisfy the same eight rules as for vectors, because each rule holds at every entry. For example,

$$(c(A + B))_{ij} = c(a_{ij} + b_{ij}) = ca_{ij} + cb_{ij} = (cA + cB)_{ij}.$$

The **transpose** $A^\top$ is the $n \times m$ matrix with $(A^\top)_{ji} = a_{ij}$. Thus

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \end{pmatrix} \implies A^\top = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ -1 & 3 \end{pmatrix}.$$

A square matrix is **symmetric** if $A^\top = A$. Entrywise definitions give

$$(A^\top)^\top = A, \quad (A + B)^\top = A^\top + B^\top, \quad (cA)^\top = cA^\top.$$

In matrix products, vectors are columns:

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad x^\top = (x_1, \dots, x_n).$$

The tuples used in Chapter I specify these same coordinates; a row vector in a matrix product is written explicitly as a transpose.

II.2. Multiplication of matrices

For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$, define

$$AB \in \mathbb{R}^{m \times p}, \quad (AB)_{ij} = \sum_{k=1}^n a_{ik}b_{kj}.$$

The shared size n is required: an entry of AB is the dot product of a row of A and a column of B .

Example 1.6. For

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 1 & -1 \\ 0 & 2 \end{pmatrix},$$

$$AB = \begin{pmatrix} 1(2) + 2(1) - 1(0) & 1(0) + 2(-1) - 1(2) \\ 0(2) + 1(1) + 3(0) & 0(0) + 1(-1) + 3(2) \end{pmatrix} = \begin{pmatrix} 4 & -4 \\ 1 & 5 \end{pmatrix}.$$

If $a_1, \dots, a_n$ are the columns of A , the entry formula also gives

$$Ax = x_1a_1 + \dots + x_na_n, \quad AB = \begin{pmatrix} Ab_1 & \dots & Ab_p \end{pmatrix},$$

where b_j is column j of B . For Example 1.6, the first column of AB is

$$2a_1 + a_2 = 2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}.$$

In particular,

$$Ax = b \iff \sum_{j=1}^n a_{ij}x_j = b_i \quad (i = 1, \dots, m).$$

For the matrix A above, $Ax = (4, 1)^\top$ means $x_1 + 2x_2 - x_3 = 4$ and $x_2 + 3x_3 = 1$.

For column vectors $x, y \in \mathbb{R}^n$,

$$x^\top y = x \cdot y \quad \text{is a scalar,} \quad xy^\top = (x_i y_j)_{ij} \quad \text{is an } n \times n \text{ matrix.}$$

Product laws. Whenever the products and sums have compatible sizes,

$$\begin{aligned} A(B + C) &= AB + AC, & (A + B)C &= AC + BC, \\ (cA)B &= A(cB) = c(AB), & (AB)C &= A(BC). \end{aligned}$$

Distributivity and scalar compatibility follow by distributing each entry's sum. More explicitly, let r_i be row i of A , and let b_j, c_j be column j of B, C . Then

$$(A(B + C))_{ij} = r_i \cdot (b_j + c_j) = r_i \cdot b_j + r_i \cdot c_j = (AB + AC)_{ij}.$$

The equality $r_i \cdot (cb_j) = c(r_i \cdot b_j)$ proves scalar compatibility. Applying the same calculation to a sum of rows proves $(A + B)C = AC + BC$.

Associativity proof (reading). If A, B, C have sizes $m \times n$, $n \times p$, $p \times q$, respectively, both sides have entry

$$\begin{aligned} ((AB)C)_{ij} &= \sum_{k=1}^p (AB)_{ik} c_{kj} = \sum_{k=1}^p \left(\sum_{\ell=1}^n a_{i\ell} b_{\ell k} \right) c_{kj} \\ &= \sum_{\ell=1}^n a_{i\ell} \left(\sum_{k=1}^p b_{\ell k} c_{kj} \right) = \sum_{\ell=1}^n a_{i\ell} (BC)_{\ell j} = (A(BC))_{ij}. \end{aligned}$$

The interchange is valid because both sums are finite. Associativity permits writing ABC .

The **identity matrix** I_n has diagonal entries 1 and all other entries 0. For $A \in \mathbb{R}^{m \times n}$, the entry formula gives

$$I_m A = A = A I_n.$$

Indeed, column j of $A I_n$ is $A e_j$, the j th column of A , while row i of $I_m A$ is row i of A . The matrix $c I_n$ is called a **scalar matrix**. For square A , define $A^0 = I_n$, $A^1 = A$, and $A^{k+1} = A^k A$. Associativity gives

$$A^r A^s = A^{r+s} \quad (r, s \geq 0).$$

Example 1.7 (Population transitions). Let x_k, y_k, z_k be the populations of Los Angeles, Chicago, and Boston in year k . Each year the fractions moving are

from\to	Los Angeles	Chicago	Boston
Los Angeles	remain	1/7	1/4
Chicago	1/5	remain	1/3
Boston	1/6	1/8	remain

The fractions remaining are $1 - 1/7 - 1/4 = 17/28$, $1 - 1/5 - 1/3 = 7/15$, and $1 - 1/6 - 1/8 = 17/24$. Thus

$$\begin{aligned} x_{k+1} &= \frac{17}{28}x_k + \frac{1}{5}y_k + \frac{1}{6}z_k, \\ y_{k+1} &= \frac{1}{7}x_k + \frac{7}{15}y_k + \frac{1}{8}z_k, \\ z_{k+1} &= \frac{1}{4}x_k + \frac{1}{3}y_k + \frac{17}{24}z_k. \end{aligned}$$

Writing populations as a column gives

$$X_{k+1} = A X_k, \quad A = \begin{pmatrix} 17/28 & 1/5 & 1/6 \\ 1/7 & 7/15 & 1/8 \\ 1/4 & 1/3 & 17/24 \end{pmatrix}, \quad X_k = \begin{pmatrix} x_k \\ y_k \\ z_k \end{pmatrix}.$$

Column j records where the population of city j goes. Each column sums to 1, so adding the three recurrence equations gives

$$x_{k+1} + y_{k+1} + z_{k+1} = x_k + y_k + z_k.$$

A square matrix with nonnegative entries and column sums 1 is a **Markov matrix** under this column-vector convention. Repeated application gives

$$X_2 = A X_1, \quad X_3 = A(A X_1) = A^2 X_1, \quad X_{k+1} = A^k X_1.$$

Proposition 1.7 (Transpose of a product). If AB is defined, then $(AB)^\top = B^\top A^\top$.

Proof. For $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, both matrices on the right and left have size $p \times m$, and

$$\left((AB)^\top\right)_{ij} = (AB)_{ji} = \sum_{k=1}^n a_{jk} b_{ki} = \sum_{k=1}^n (B^\top)_{ik} (A^\top)_{kj} = (B^\top A^\top)_{ij}.$$

□

Consequently $Ax = b$ can also be written $x^\top A^\top = b^\top$, with the order reversed.

Example 1.8. Let

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then $PN = N$, whereas $NP = 0$. Thus matrix multiplication need not commute, and a product of two nonzero matrices may be zero. Also $P^2 = PI_2 = P$, although $P \neq I_2$; cancelling a nonzero matrix from an equation is not valid in general.

Inverses. An **inverse** of a square matrix $A \in \mathbb{R}^{n \times n}$ is an $n \times n$ matrix B satisfying both $AB = I_n$ and $BA = I_n$. A matrix with an inverse is **invertible**. The inverse is unique: if B and C are inverses, then

$$B = BI_n = B(AC) = (BA)C = I_n C = C.$$

We denote it by A^{-1} .

For example,

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \implies AB = BA = I_2,$$

so $B = A^{-1}$. In contrast, the matrix P in Example 1.8 has no inverse: every product PC has zero second row and therefore cannot equal I_2 .

If A is invertible and $AX = AY$, multiplication on the left by A^{-1} gives $X = Y$. This cancellation uses invertibility, not merely $A \neq 0$.

Transposing the two inverse identities gives

$$(A^{-1})^\top A^\top = (AA^{-1})^\top = I_n, \quad A^\top (A^{-1})^\top = (A^{-1}A)^\top = I_n.$$

Thus $A^\top$ is invertible and $(A^\top)^{-1} = (A^{-1})^\top$.

Example 1.9 (Rotations). For $\theta \in \mathbb{R}$, put

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

A unit vector in the plane has the form $u = (\cos \varphi, \sin \varphi)^\top$. Multiplication gives

$$R(\theta)u = \begin{pmatrix} \cos \theta \cos \varphi - \sin \theta \sin \varphi \\ \sin \theta \cos \varphi + \cos \theta \sin \varphi \end{pmatrix} = \begin{pmatrix} \cos(\theta + \varphi) \\ \sin(\theta + \varphi) \end{pmatrix}.$$

The last equality is the pair of trigonometric addition formulas. For any nonzero $x = ru$, where $r = \|x\|$, $R(\theta)x = rR(\theta)u$: multiplication preserves length and turns the direction through θ . It fixes the zero vector. For example, rotating $(2, 5)^\top$ through $\pi/3$ gives

$$\begin{pmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 - 5\sqrt{3}/2 \\ \sqrt{3} + 5/2 \end{pmatrix}.$$

If coordinates are written as a row, the same rotation is $x^\top R(\theta)^\top$, by the transpose-product rule.

II.3. Homogeneous linear equations and elimination

For $A = (a_{ij}) \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and an unknown column $x \in \mathbb{R}^n$, the system

$$\sum_{j=1}^n a_{ij}x_j = b_i \quad (1 \leq i \leq m) \quad \text{is written} \quad Ax = b.$$

It has m equations and n unknowns. A **solution** is a column whose entries satisfy every equation. The system is **consistent** if it has a solution, and **homogeneous** if $b = 0$. The homogeneous system always has the **trivial solution** $x = 0$; a **nontrivial solution** has at least one nonzero coordinate. If the rows of A are $r_1^\top, \dots, r_m^\top$, then

$$Ax = 0 \quad \Longleftrightarrow \quad r_i \cdot x = 0 \text{ for every } i.$$

Example 1.10 (Eliminate one variable, then recover it). Consider

$$2x + 3y - z = 0, \quad x + y + z = 0.$$

Subtracting twice the second equation from the first gives $y - 3z = 0$. Set $z = t$. Then $y = 3t$, and the second original equation gives $x = -y - z = -4t$. The first original equation is also satisfied, since

$$2x + 3y - z = 2(x + y + z) + (y - 3z) = 0.$$

Thus every $t(-4, 3, 1)$ is a solution, and every solution is obtained by taking $t = z$. Choosing $t = 1$ gives a nontrivial solution.

Example 1.11. Let

$$A = \begin{pmatrix} 1 & 2 & -1 & 1 \\ 2 & 4 & -1 & 3 \\ 3 & 6 & -2 & 4 \end{pmatrix}.$$

The third equation of $Ax = 0$ is the sum of the first two. Subtracting twice the first equation from the second leaves the equivalent system

$$x_1 + 2x_2 - x_3 + x_4 = 0, \quad x_3 + x_4 = 0.$$

Choose $x_2 = s$, $x_4 = t$. Solving backwards gives $x_3 = -t$, $x_1 = -2s - 2t$. Thus all solutions are

$$x = \begin{pmatrix} -2s - 2t \\ s \\ -t \\ t \end{pmatrix} = s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -2 \\ 0 \\ -1 \\ 1 \end{pmatrix}, \quad s, t \in \mathbb{R}.$$

Every pair (s, t) gives a solution, and every solution has this form because its second and fourth coordinates determine s, t . The solution is nontrivial exactly when $(s, t) \neq (0, 0)$.

Theorem 1.8 (More unknowns than equations). If $A \in \mathbb{R}^{m \times n}$ and $n > m$, then $Ax = 0$ has a nontrivial solution.

Proof. We induct on the number m of equations. For one equation

$$a_1x_1 + \cdots + a_nx_n = 0, \quad n > 1,$$

if all coefficients vanish, $x = e_1$ is a nontrivial solution. Otherwise renumber the variables so that $a_1 \neq 0$. Set $x_2 = \cdots = x_n = 1$, and then set

$$x_1 = -\frac{a_2 + \cdots + a_n}{a_1}.$$

This solves the equation and is nontrivial because $x_2 = 1$.

Suppose the claim holds for $m - 1$ equations in more than $m - 1$ unknowns. If $A = 0$, again use e_1 . Otherwise renumber equations and variables so that $a_{11} \neq 0$. For each $i = 2, \dots, m$, subtract a_{i1}/a_{11} times the first equation from equation i . Writing $r_i^\top$ for row i , the new equations are

$$\left(r_i - \frac{a_{i1}}{a_{11}}r_1\right) \cdot x = 0, \quad i = 2, \dots, m.$$

The coefficient of x_1 is $a_{i1} - (a_{i1}/a_{11})a_{11} = 0$. Hence these equations are

$$\sum_{j=2}^n \left(a_{ij} - \frac{a_{i1}}{a_{11}}a_{1j}\right) x_j = 0, \quad i = 2, \dots, m,$$

a system of $m - 1$ equations in $n - 1 > m - 1$ unknowns. By induction, choose a solution $(x_2, \dots, x_n) \neq 0$. Recover the first coordinate from the first original equation:

$$x_1 = -\frac{a_{12}x_2 + \dots + a_{1n}x_n}{a_{11}}.$$

Now $r_1 \cdot x = 0$. For every remaining original row,

$$r_i \cdot x = \left(r_i - \frac{a_{i1}}{a_{11}}r_1\right) \cdot x + \frac{a_{i1}}{a_{11}}(r_1 \cdot x) = 0.$$

Thus $Ax = 0$, and $x \neq 0$ because its last $n - 1$ coordinates were chosen not all zero. This completes the induction. $\square$

II.4. Row operations and Gauss elimination

The **augmented matrix** $[A \mid b]$ records the coefficients and the right-hand side of $Ax = b$. The **elementary row operations** are

$$\begin{array}{lll} \text{Interchange:} & R_i \leftrightarrow R_j, & i \neq j, \\ \text{Scaling:} & R_i \leftarrow cR_i, & c \neq 0, \\ \text{Replacement:} & R_i \leftarrow R_i + cR_j, & i \neq j. \end{array}$$

Two matrices are **row equivalent**, written $M \sim N$, if a finite sequence of these operations changes one into the other. The sequence may be empty.

Proposition 1.9. Row operations on the augmented matrix preserve exactly the solutions of a linear system.

Proof. Interchanging equations only changes their order. Multiplying an equation by $c \neq 0$ preserves its solutions, since division by c recovers it. For replacement, the two old equations

$$r_i \cdot x = b_i, \quad r_j \cdot x = b_j$$

imply

$$(r_i + cr_j) \cdot x = b_i + cb_j, \quad r_j \cdot x = b_j.$$

Conversely, subtracting c times the unchanged second equation from the new first equation recovers $r_i \cdot x = b_i$. Thus each operation preserves the solution set in both directions, as does a sequence of them. $\square$

The operation acts on the *entire* row, including its right-hand side. Multiplication by zero is excluded because it can discard an equation. In a replacement, row j remains unchanged.

A matrix is in **row echelon form** if its zero rows are at the bottom and the first nonzero entry of each successive nonzero row occurs strictly farther to the right. These first nonzero entries are the **pivots**; their columns are **pivot columns**. Pivots need not equal 1. For example,

$$\begin{pmatrix} 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

has pivots in columns 2 and 4.

Theorem 1.10 (Gauss elimination). Every matrix is row equivalent to a matrix in row echelon form.

Proof. If the matrix is zero, it is already in echelon form. Otherwise choose the leftmost nonzero column, say column j , and interchange rows to put a nonzero entry p in the first row. The matrix now has the form

$$\begin{pmatrix} 0 & \cdots & 0 & p & b_{1,j+1} & \cdots & b_{1n} \\ 0 & \cdots & 0 & b_{2j} & b_{2,j+1} & \cdots & b_{2n} \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & \cdots & 0 & b_{mj} & b_{m,j+1} & \cdots & b_{mn} \end{pmatrix}, \quad p \neq 0.$$

The initial $j - 1$ columns are zero. Apply $R_i \leftarrow R_i - (b_{ij}/p)R_1$ for $i = 2, \dots, m$. In column j this gives $b_{ij} - (b_{ij}/p)p = 0$; in a later column k it gives $b'_{ik} = b_{ik} - (b_{ij}/p)b_{1k}$. Thus

$$\begin{pmatrix} 0 & \cdots & 0 & p & b_{1,j+1} & \cdots & b_{1n} \\ 0 & \cdots & 0 & 0 & b'_{2,j+1} & \cdots & b'_{2n} \\ \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & \cdots & 0 & 0 & b'_{m,j+1} & \cdots & b'_{mn} \end{pmatrix}$$

is row equivalent to the original matrix. Repeat the procedure on the smaller matrix

$$B' = \begin{pmatrix} b'_{2,j+1} & \cdots & b'_{2n} \\ \vdots & & \vdots \\ b'_{m,j+1} & \cdots & b'_{mn} \end{pmatrix}.$$

Operations among these lower rows preserve all their earlier zero entries. Each stage fixes a new pivot strictly to the right of the previous pivot and leaves fewer rows and columns to process. If the remaining matrix is zero or has no rows or columns, stop. The fixed pivot rows followed by any zero rows form an echelon matrix. $\square$

For a system, apply this procedure to the coefficient columns and carry the right-hand side along. Write the result as $[U \mid d]$, with U in echelon form. If U has r nonzero rows, their pivot columns $j_1 < \cdots < j_r$ identify the **pivot variables** $x_{j_1}, \dots, x_{j_r}$. The other variables are **free**.

Proposition 1.11 (Consistency and free variables). The system is consistent if and only if every zero row of U has corresponding entry zero in d . If consistent, every choice of the $n - r$ free variables determines exactly one solution. A consistent system has a unique solution if and only if there are no free variables.

Proof. A row $[0 \cdots 0 \mid d_i]$ requires $0 = d_i$. If these requirements hold, assign arbitrary values to the free variables and solve the nonzero rows from bottom to top. In row i ,

$$x_{j_i} = \frac{d_i - \sum_{k > j_i} u_{ik} x_k}{u_{ij_i}}.$$

The denominator is nonzero. Every variable on the right is either free or already determined by a lower row, so the formula gives exactly one value. This also shows that every solution is obtained. Different choices of a free coordinate give different solutions. $\square$

Example 1.12. For the matrix A of Example 1.11, let $b = (b_1, b_2, b_3)^\top$. The operations

$$R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 - 3R_1, \quad R_3 \leftarrow R_3 - R_2$$

give

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & b_1 \\ 2 & 4 & -1 & 3 & b_2 \\ 3 & 6 & -2 & 4 & b_3 \end{array} \right] \sim \left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & b_1 \\ 0 & 0 & 1 & 1 & b_2 - 2b_1 \\ 0 & 0 & 1 & 1 & b_3 - 3b_1 \end{array} \right]$$

after the first two operations, and the last operation gives

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & b_1 \\ 0 & 0 & 1 & 1 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_1 - b_2 \end{array} \right].$$

Hence $Ax = b$ is consistent exactly when $b_3 = b_1 + b_2$. The pivot variables are x_1, x_3 ; the free variables are x_2, x_4 . When the condition holds, back substitution gives

$$x_2 = s, \quad x_4 = t, \quad x_3 = b_2 - 2b_1 - t, \quad x_1 = b_1 - 2s + x_3 - t = b_2 - b_1 - 2s - 2t.$$

Thus the complete solution is

$$x = \begin{pmatrix} b_2 - b_1 - 2s - 2t \\ s \\ b_2 - 2b_1 - t \\ t \end{pmatrix}, \quad s, t \in \mathbb{R}.$$

For $b = (1, 3, 4)^\top$, the solutions are $(2 - 2s - 2t, s, 1 - t, t)^\top$. For $b = (1, 3, 5)^\top$, there is no solution.

The same existence theorem in echelon form. We can now give a second proof of Theorem 1.8. Elimination leaves at most one pivot per row, so the number r of pivots satisfies $r \leq m < n$. Row operations leave the zero right-hand side zero; the system is consistent. At least one variable is free. Set one free variable to 1 and all other free variables to 0, then use back substitution to determine the pivot variables. The resulting solution is nonzero because its chosen free coordinate is 1.

The condition $n > m$ is sufficient, not necessary: $x_1 + x_2 = 0$ and $2x_1 + 2x_2 = 0$ have nontrivial solutions with $n = m = 2$. For a nonhomogeneous system, $n > m$ does not ensure consistency, as Example 1.12 shows.

Exercises and solutions

Worked in class: 1.24(a).

Suggested practice: 1.2, 1.5, 1.10, 1.13, 1.19, 1.22, 1.24, 1.26, 1.30, 1.31, 1.32.

Dot products and norms

Exercise 1.1. Give vectors x, y, z for which $x \perp y$ and $y \perp z$, but x is not perpendicular to z .

Solution. Take $x = z = (1, 0)$ and $y = (0, 1)$. Then $x \cdot y = y \cdot z = 0$, whereas $x \cdot z = 1$.

Exercise 1.2. For $x, y \in \mathbb{R}^n$, prove

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2, \quad \|x + y\|^2 - \|x - y\|^2 = 4x \cdot y.$$

Interpret the first identity as a parallelogram law.

Solution. The dot-product laws give

$$\|x + y\|^2 = \|x\|^2 + 2x \cdot y + \|y\|^2, \quad \|x - y\|^2 = \|x\|^2 - 2x \cdot y + \|y\|^2.$$

Add and subtract these equations. The parallelogram with vertices $0, x, x + y, y$ has diagonals of lengths $\|x + y\|$ and $\|x - y\|$. The first identity says that the sum of their squared lengths equals the sum of the squared lengths of the four sides.

Exercise 1.3. Give three nonzero vectors a, b, c such that $a \cdot b = a \cdot c$ but $b \neq c$.

Solution. Take $a = (1, 0)$, $b = (1, 1)$, and $c = (1, 2)$. Both dot products are 1, although $b \neq c$.

Exercise 1.4. Let $u_1, \dots, u_r \in \mathbb{R}^n$ be nonzero and pairwise perpendicular. If $c_1 u_1 + \dots + c_r u_r = 0$, prove that $c_1 = \dots = c_r = 0$.

Solution. For each j , take the dot product with u_j :

$$0 = \sum_{i=1}^r c_i (u_i \cdot u_j) = c_j \|u_j\|^2.$$

Since $u_j \neq 0$, its squared norm is positive, so $c_j = 0$.

Exercise 1.5. For $a = (2, -1, 5)$ and $b = (-1, 1, 1)$, find the projection of a along b and the projection of b along a .

Solution. Here $a \cdot b = -2 - 1 + 5 = 2$, $\|a\|^2 = 30$, and $\|b\|^2 = 3$. Thus the projections are, respectively,

$$\frac{a \cdot b}{\|b\|^2} b = \left(-\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right), \quad \frac{b \cdot a}{\|a\|^2} a = \left(\frac{2}{15}, -\frac{1}{15}, \frac{1}{3}\right).$$

Exercise 1.6. Use the Schwarz inequality to prove that, for $x, y \in \mathbb{R}^n$,

$$\|x + y\| \leq \|x\| + \|y\|.$$

Prove that equality holds exactly when $y = 0$ or $x = ky$ for some $k \geq 0$.

Solution. By Schwarz,

$$\|x + y\|^2 = \|x\|^2 + 2x \cdot y + \|y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2.$$

Taking nonnegative square roots proves the inequality. Equality requires $x \cdot y = \|x\|\|y\|$. If $y = 0$, this holds. If $y \neq 0$, the equality condition in Schwarz requires $x = ky$ for a real k . Then $x \cdot y = k\|y\|^2$ and $\|x\|\|y\| = |k|\|y\|^2$, which are equal exactly when $k \geq 0$.

Lines and planes

Exercise 1.7. Let $x \neq y$ and $0 \leq \lambda \leq 1$. Show that $z = (1 - \lambda)x + \lambda y$ divides the segment from x to y in the proportion $\lambda : (1 - \lambda)$.

Solution. The representation $z = x + \lambda(y - x)$ places z on the segment. Moreover,

$$\|z - x\| = \|\lambda(y - x)\| = \lambda\|y - x\|, \quad \|y - z\| = (1 - \lambda)\|y - x\|.$$

Since $\|y - x\| > 0$, these distances have the stated proportion, including $z = x$ at $\lambda = 0$ and $z = y$ at $\lambda = 1$.

Exercise 1.8. Find the equation of the plane perpendicular to $(1, -1, 3)$ and passing through $(4, 2, -1)$.

Solution. A point (x_1, x_2, x_3) lies on the plane exactly when

$$(1, -1, 3) \cdot (x_1 - 4, x_2 - 2, x_3 + 1) = 0.$$

Thus its equation is $x_1 - x_2 + 3x_3 = -1$.

Exercise 1.9. Write down vectors perpendicular to the lines $y = mx + b$ and $y = m'x + c$. Show that these vectors are perpendicular to each other exactly when $mm' = -1$.

Solution. The equations $-mx + y = b$ and $-m'x + y = c$ give nonzero normal vectors $(-m, 1)$ and $(-m', 1)$. Their dot product is $mm' + 1$, which vanishes exactly when $mm' = -1$.

Exercise 1.10. (a) Find the intersection of the line through $(1, 3, 5)$ with direction $(-2, 1, 1)$ and the plane $2x + 3y - z = 1$.

(b) Find the intersection of the plane $3x - 4y + z = 2$ and the line through $(1, 2, -1)$ perpendicular to that plane.

Solution. (a) Substitute $(x, y, z) = (1 - 2t, 3 + t, 5 + t)$ into the plane equation:

$$2(1 - 2t) + 3(3 + t) - (5 + t) = 6 - 2t = 1.$$

Hence $t = 5/2$ and the intersection is $(-4, 11/2, 15/2)$.

(b) The normal vector $(3, -4, 1)$ gives the line $(x, y, z) = (1 + 3t, 2 - 4t, -1 + t)$. Substitution gives

$$3(1 + 3t) - 4(2 - 4t) + (-1 + t) = -6 + 26t = 2.$$

Thus $t = 4/13$ and the intersection is $(25/13, 10/13, -9/13)$.

Matrix multiplication and transpose

Exercise 1.11. Let

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 3 & 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 2 & 0 \\ 3 & -1 \end{pmatrix}, \quad c = \begin{pmatrix} 1 \\ 3 \end{pmatrix}.$$

Compute $(AB)c$ and $A(Bc)$, showing the intermediate products.

Solution. The intermediate products have sizes 2×2 and 3×1 :

$$AB = \begin{pmatrix} 2+2-3 & 2+1 \\ 3+2+6 & 3-2 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 11 & 1 \end{pmatrix}, \quad Bc = \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix}.$$

Consequently,

$$(AB)c = \begin{pmatrix} 1+9 \\ 11+3 \end{pmatrix} = \begin{pmatrix} 10 \\ 14 \end{pmatrix}, \quad A(Bc) = \begin{pmatrix} 8+2 \\ 12+2 \end{pmatrix} = \begin{pmatrix} 10 \\ 14 \end{pmatrix}.$$

Exercise 1.12. On an ordered triple (a_1, a_2, a_3) , let S reverse the order and let T move the last entry to the first position. Show that performing S then T need not give the same result as performing T then S . Do this first with triples, then with matrices acting on column vectors.

Solution. Applying S then T gives $(a_1, a_2, a_3) \mapsto (a_3, a_2, a_1) \mapsto (a_1, a_3, a_2)$. Applying T then S gives $(a_1, a_2, a_3) \mapsto (a_3, a_1, a_2) \mapsto (a_2, a_1, a_3)$. For instance, the triple $(1, 2, 3)$ gives different results. The corresponding matrices are

$$S = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

The two orders give

$$TS = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad ST = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which are unequal.

Exercise 1.13. Construct real 2×2 matrices satisfying each condition:

- (a) $A^2 = -I$;
- (b) $B^2 = 0$ but $B \neq 0$;
- (c) $CD = -DC$ and $CD \neq 0$;
- (d) $EF = 0$, with every entry of both E and F nonzero.

Solution. (a) Take $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Then $A^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I$.

(b) Take $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. It is nonzero, and direct multiplication gives $B^2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

(c) Take

$$C = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then $CD = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $DC = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -CD$.

(d) Take $E = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $F = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$. Every entry is nonzero, but each entry of EF is either $1 - 1$ or $-1 + 1$, so $EF = 0$.

Exercise 1.14. Two square matrices A, B of the same size *commute* if $AB = BA$. Prove that

$$(A + B)(A - B) = A^2 - B^2$$

holds exactly when A and B commute.

Solution. Distributivity gives

$$(A + B)(A - B) = A^2 - AB + BA - B^2.$$

This equals $A^2 - B^2$ exactly when $-AB + BA = 0$, or $AB = BA$.

Exercise 1.15. (a) For symmetric $n \times n$ matrices A, B , prove that AB is symmetric exactly when $AB = BA$.

(b) Give two symmetric matrices whose product is not symmetric.

Solution. (a) Since $A^\top = A$ and $B^\top = B$,

$$(AB)^\top = B^\top A^\top = BA.$$

Thus $(AB)^\top = AB$ exactly when $BA = AB$.

(b) The symmetric matrices $A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ give $AB = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}$, which is not symmetric.

Exercise 1.16. A real square matrix K is *skew-symmetric* if $K^\top = -K$. Let A be a real square matrix.

(a) Prove that $A - A^\top$ is skew-symmetric.

(b) Express A as the sum of a symmetric matrix and a skew-symmetric matrix.

(c) Prove that every diagonal entry of a skew-symmetric matrix is zero.

Solution. (a) Entrywise transposition gives $(A - A^\top)^\top = A^\top - A = -(A - A^\top)$.

(b) Set

$$H = \frac{A + A^\top}{2}, \quad K = \frac{A - A^\top}{2}.$$

Then $A = H + K$, $H^\top = H$, and $K^\top = -K$.

(c) If $K^\top = -K$, the (i, i) entry gives $k_{ii} = -k_{ii}$. Hence $2k_{ii} = 0$, so $k_{ii} = 0$.

Exercise 1.17. For real columns a, b of the same length, determine exactly when $ab^\top = ba^\top$.

Solution. If $b = 0$, both matrices are zero. Suppose $b \neq 0$ and choose an index j with $b_j \neq 0$. Equality of the (i, j) entries gives $a_i b_j = b_i a_j$ for every i , so

$$a = \frac{a_j}{b_j} b.$$

Conversely, if $a = kb$ for a real k , then $ab^\top = kbb^\top = ba^\top$. The exact condition is therefore $b = 0$ or $a = kb$ for some real k ; this includes $a = 0$.

Exercise 1.18. Let $A = [a_1 \cdots a_n]$ be an $m \times n$ real matrix. Prove

$$AA^\top = \sum_{j=1}^n a_j a_j^\top, \quad (A^\top A)_{ij} = a_i \cdot a_j.$$

Solution. Write the entries of A as a_{rj} . The (r, s) entry of the first product is

$$(AA^\top)_{rs} = \sum_{j=1}^n a_{rj} a_{sj} = \sum_{j=1}^n (a_j a_j^\top)_{rs}.$$

For the second product,

$$(A^\top A)_{ij} = \sum_{r=1}^m a_{ri} a_{rj} = a_i \cdot a_j.$$

Both identities follow by equality of entries.

Exercise 1.19. Let $A \in \mathbb{R}^{m \times n}$ and $B, C \in \mathbb{R}^{n \times p}$. Prove:

- (a) $A^\top A = 0$ if and only if $A = 0$;
- (b) $AB = 0$ if and only if $A^\top AB = 0$;
- (c) $AB = AC$ if and only if $A^\top AB = A^\top AC$.

Solution. (a) If $A^\top A = 0$, then, for every column j ,

$$0 = (A^\top A)_{jj} = \sum_{i=1}^m a_{ij}^2.$$

A sum of squares of real numbers is zero only when every summand is zero. Thus all entries of A vanish. The converse follows by multiplication.

(b) The forward implication follows by multiplying $AB = 0$ on the left by $A^\top$. Conversely, if $A^\top AB = 0$, then

$$(AB)^\top (AB) = B^\top A^\top AB = 0.$$

Part (a), applied to AB , gives $AB = 0$.

(c) Apply part (b) with $B - C$ in place of B :

$$AB = AC \iff A(B - C) = 0 \iff A^\top A(B - C) = 0 \iff A^\top AB = A^\top AC.$$

Exercise 1.20. For a square matrix M , define its *trace* by $\text{tr}(M) = \sum_i M_{ii}$.

(a) If A, B are real $m \times n$ matrices, prove

$$\text{tr}(A^\top B) = \text{tr}(BA^\top) = \text{tr}(AB^\top) = \text{tr}(B^\top A).$$

(b) Prove $\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA)$ and specify the compatible sizes of A, B, C .

(c) Must $\text{tr}(ABC) = \text{tr}(ACB)$ whenever both sides are defined?

Solution. (a) If the entries of A, B are a_{ij}, b_{ij} , then

$$\text{tr}(A^\top B) = \sum_{j=1}^n \sum_{i=1}^m a_{ij} b_{ij}, \quad \text{tr}(BA^\top) = \sum_{i=1}^m \sum_{j=1}^n b_{ij} a_{ij}.$$

These are equal because finite sums can be reordered and real multiplication commutes. Interchanging A and B gives the other two traces, with the same sum.

(b) Take A of size $m \times n$, B of size $n \times p$, and C of size $p \times m$. Then ABC, CAB, BCA are square matrices of sizes m, p, n , respectively. For U of size $r \times s$ and V of size $s \times r$, part (a) applied to $U^\top, V$ gives $\text{tr}(UV) = \text{tr}(VU)$. Hence

$$\text{tr}(ABC) = \text{tr}((AB)C) = \text{tr}(C(AB)) = \text{tr}(CAB),$$

and, applying the same rule to CA and B , $\text{tr}(CAB) = \text{tr}(BCA)$.

(c) No. Take

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Here $BC = A$, whereas $CB = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. Thus $ABC = A$ and $ACB = 0$, giving traces 1 and 0.

Inverses and repeated multiplication

Exercise 1.21. Let A, B be invertible real $n \times n$ matrices. Prove that AB and $A^\top$ are invertible, with

$$(AB)^{-1} = B^{-1}A^{-1}, \quad (A^\top)^{-1} = (A^{-1})^\top.$$

Solution. Associativity gives both required products:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I,$$

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I.$$

For the transpose, the product rule gives

$$A^\top(A^{-1})^\top = (A^{-1}A)^\top = I, \quad (A^{-1})^\top A^\top = (AA^{-1})^\top = I.$$

Each displayed candidate is a two-sided inverse, so uniqueness of the inverse proves the formulas.

Exercise 1.22. Let A be a real $n \times n$ matrix. In each case, prove that the indicated matrix is invertible by finding its inverse:

(a) $I - A$, if $A^k = 0$ for a positive integer k ;

(b) A , if $A^2 + 2A + I = 0$;

(c) A , if $A^2 - A + I = 0$.

Here $A^0 = I$ and A^j means a product of j copies of A for $j \geq 1$.

Solution. (a) Put $S = \sum_{j=0}^{k-1} A^j$. Distributivity and associativity give

$$(I - A)S = S(I - A) = \sum_{j=0}^{k-1} A^j - \sum_{j=1}^k A^j = I - A^k = I.$$

Thus $(I - A)^{-1} = S$. When $k = 1$, the hypothesis says $A = 0$ and $S = I$, so the calculation includes that case.

(b) The hypothesis gives

$$A(-A - 2I) = (-A - 2I)A = -A^2 - 2A = I.$$

Hence $A^{-1} = -A - 2I$.

(c) The hypothesis gives

$$A(I - A) = (I - A)A = A - A^2 = I,$$

so $A^{-1} = I - A$.

Exercise 1.23. Let $n \geq 1$, let $j = (1, \dots, 1)^\top \in \mathbb{R}^n$, and set

$$P = I_n - \frac{1}{n}jj^\top.$$

Prove that P is symmetric and *idempotent*, meaning $P^2 = P$. Explain these properties by describing Px for a column $x \in \mathbb{R}^n$.

Solution. Since $j^\top j = n$, associativity gives

$$(jj^\top)^2 = j(j^\top j)j^\top = njj^\top.$$

Therefore

$$P^2 = I_n - \frac{2}{n}jj^\top + \frac{1}{n^2}(jj^\top)^2 = I_n - \frac{1}{n}jj^\top = P.$$

Also $(jj^\top)^\top = jj^\top$, so $P^\top = P$. If $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$, then

$$Px = x - \bar{x}j.$$

Thus P subtracts the mean from each coordinate. The result has coordinate sum $\sum_i (x_i - \bar{x}) = 0$, so applying P again changes nothing. Its diagonal entries are $1 - 1/n$ and its off-diagonal entries are $-1/n$: each coordinate contributes the same amount to every other coordinate, giving symmetry.

Homogeneous systems and solution sets

Exercise 1.24. For each system, write its coefficient matrix, reduce it to row echelon form, and find all solutions.

(a) $2x - 3y + 4z = 0, 3x + y + z = 0.$

(b) $-x + 2y - 4z + w = 0, x + 3y + z - w = 0.$

Solution. (a) The coefficient matrix and a row reduction are

$$\begin{pmatrix} 2 & -3 & 4 \\ 3 & 1 & 1 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - \frac{3}{2}R_1} \begin{pmatrix} 2 & -3 & 4 \\ 0 & 11/2 & -5 \end{pmatrix}.$$

The second row gives $y = 10z/11$. Substituting into the first gives

$$2x = 3y - 4z = \frac{30}{11}z - 4z = -\frac{14}{11}z, \quad x = -\frac{7}{11}z.$$

Writing $z = 11t$, all solutions are $t(-7, 10, 11)^\top$, $t \in \mathbb{R}$.

(b) Here

$$\begin{pmatrix} -1 & 2 & -4 & 1 \\ 1 & 3 & 1 & -1 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_1} \begin{pmatrix} -1 & 2 & -4 & 1 \\ 0 & 5 & -3 & 0 \end{pmatrix}.$$

Thus $y = 3z/5$, and the first row gives $x = 2y - 4z + w = 6z/5 - 4z + w = w - 14z/5$. With $z = 5s$ and $w = t$, all solutions are

$$(x, y, z, w)^\top = (-14s + t, 3s, 5s, t)^\top, \quad s, t \in \mathbb{R}.$$

In both parts the free coordinates can be chosen arbitrarily, and back substitution determines the remaining coordinates. Hence the displayed sets include every solution; substituting the formulas verifies them.

Exercise 1.25. Let $a_1, \dots, a_m \in \mathbb{R}^n$. Suppose $x \cdot a_i = y \cdot a_i = 0$ for every i . Prove that $x + y$ also satisfies these equations and that cx satisfies them for every real number c .

Solution. For each i , distributivity gives $(x + y) \cdot a_i = x \cdot a_i + y \cdot a_i = 0$. Scalar compatibility gives $(cx) \cdot a_i = c(x \cdot a_i) = 0$. These equalities hold for every row, proving both assertions.

Exercise 1.26. Let $A \in \mathbb{R}^{m \times n}$ and suppose $Ax = b$. If $Ax' = b$ as well, prove that $x' = x + y$ for some y satisfying $Ay = 0$. Conversely, show that $x + y$ solves the original system whenever $Ay = 0$.

Solution. Set $y = x' - x$. Then $Ay = Ax' - Ax = b - b = 0$ and $x' = x + y$. Conversely, $A(x + y) = Ax + Ay = b + 0 = b$. Thus the full solution set is one particular solution plus all solutions of the homogeneous system.

Exercise 1.27. Let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{m \times p}$, and suppose $AX_i = B$ for $i = 1, \dots, k$, where every X_i is $n \times p$. If real numbers $c_1, \dots, c_k$ satisfy $\sum_i c_i = 1$, prove that $\sum_i c_i X_i$ also solves $AX = B$.

Solution. All matrices in the sum have the same size. Distributivity gives

$$A \left(\sum_{i=1}^k c_i X_i \right) = \sum_{i=1}^k c_i A X_i = \sum_{i=1}^k c_i B = \left(\sum_{i=1}^k c_i \right) B = B.$$

Thus the asserted matrix is a solution, including when $B = 0$.

Consistency and exceptional parameters

Exercise 1.28. For which real numbers k does the system

$$x_1 + x_2 = 1, \quad 2x_1 + 2x_2 = k$$

have a solution? When it has a solution, is that solution unique?

Solution. Subtracting twice the first equation from the second gives $0 = k - 2$. Thus a solution exists exactly when $k = 2$. In that case the second equation repeats the first, and all solutions are $(1 - t, t)^\top$, $t \in \mathbb{R}$. There are infinitely many, so the solution is not unique.

Exercise 1.29. Can a matrix A be chosen so that $Ax = b$ is inconsistent for every right-hand side b of the appropriate size? Can a matrix A be chosen so that $Ax = b$ is consistent for every such b ? Justify each answer.

Solution. The first answer is no: for every matrix A , the right-hand side $b = 0$ has the solution $x = 0$. The second answer is yes: choose $A = I_m$. Then $x = b$ solves $Ax = b$ for every $b \in \mathbb{R}^m$.

Exercise 1.30. Does the following system have a solution?

$$\begin{aligned} 2x_1 + 3x_2 + x_3 &= 7, \\ x_1 + 2x_2 - 2x_3 &= 8, \\ 3x_1 + 4x_2 + 4x_3 &= 20. \end{aligned}$$

Solution. The coefficient rows satisfy

$$2(2, 3, 1) - (1, 2, -2) = (3, 4, 4).$$

Thus twice the first left-hand side minus the second equals the third left-hand side. The same operation on the right-hand sides gives $2(7) - 8 = 6$, not 20. Subtracting twice row one and adding row two to row three therefore produces $0 = 14$. The system is inconsistent.

Exercise 1.31. For real parameters α, β , consider

$$\begin{aligned}\alpha x_1 + \beta x_2 + 2x_3 &= 1, \\ \alpha x_1 + (2\beta - 1)x_2 + 3x_3 &= 1, \\ \alpha x_1 + \beta x_2 + (\beta + 3)x_3 &= 2\beta - 1.\end{aligned}$$

When does a solution exist, and when is it unique?

Solution. Subtract the first equation from the second and third. The equivalent system is

$$\alpha x_1 + \beta x_2 + 2x_3 = 1, \quad (\beta - 1)x_2 + x_3 = 0, \quad (\beta + 1)x_3 = 2(\beta - 1).$$

If $\beta = -1$, the last equation is $0 = -4$, so there is no solution. If $\beta = 1$, then $x_3 = 0$ and $\alpha x_1 + x_2 = 1$; all solutions are $(t, 1 - \alpha t, 0)^\top$, $t \in \mathbb{R}$, for every α .

For $\beta \neq \pm 1$, the last equation first gives x_3 , and the second gives $x_2 = -x_3/(\beta - 1)$. Hence

$$x_3 = \frac{2(\beta - 1)}{\beta + 1}, \quad x_2 = -\frac{2}{\beta + 1}.$$

Substituting into the first equation now gives

$$\alpha x_1 = 1 - \beta x_2 - 2x_3 = 1 + \frac{2\beta}{\beta + 1} - \frac{4(\beta - 1)}{\beta + 1} = \frac{5 - \beta}{\beta + 1}.$$

If $\alpha \neq 0$, there is exactly one solution, with $x_1 = (5 - \beta)/(\alpha(\beta + 1))$. If $\alpha = 0$, consistency requires $\beta = 5$; the solutions are then $(t, -1/3, 4/3)^\top$, $t \in \mathbb{R}$. Thus uniqueness holds exactly when $\alpha \neq 0$ and $\beta \neq \pm 1$. The remaining consistent cases are $\beta = 1$ with arbitrary α , and $(\alpha, \beta) = (0, 5)$; each has infinitely many solutions. All other parameter pairs give no solution.

Point-plane distances

Exercise 1.32. Find the distance between each point and plane:

- (a) $(1, 1, 2)$ and $3x + y - 5z = 2$;
- (b) $(-1, 3, 2)$ and $2x - 4y + z = 1$;
- (c) $(3, -2, 1)$ and the yz -plane;
- (d) $(-3, -2, 1)$ and the yz -plane.

Solution. For a plane $n \cdot x = c$ with $n \neq 0$ and a point q , the distance is $|n \cdot q - c|/\|n\|$. (a) Here $n = (3, 1, -5)$, $c = 2$, and $\|n\| = \sqrt{9 + 1 + 25} = \sqrt{35}$. Thus

$$d = \frac{|3(1) + 1 - 5(2) - 2|}{\sqrt{35}} = \frac{8}{\sqrt{35}}.$$

(b) Here $n = (2, -4, 1)$, $c = 1$, and $\|n\| = \sqrt{4 + 16 + 1} = \sqrt{21}$. Hence

$$d = \frac{|2(-1) - 4(3) + 2 - 1|}{\sqrt{21}} = \frac{13}{\sqrt{21}}.$$

(c,d) The yz -plane has equation $x = 0$, with unit normal $(1, 0, 0)$. The two distances are therefore $|3| = 3$ and $|-3| = 3$. The points lie on opposite sides of the plane; the absolute value makes both distances nonnegative.